

Q1**(6 marks)**

Sampling replica tones at 7, 13, 27, 33, 43 MHz is shaped by sinc function. Sinc zeros at $n \cdot (f_s) = 20, 40, 60$ MHz

Time domain convolution of samples with ZOH is Frequency domain multiplication

$$H(f) = (\tau/T_s) \text{sinc}(\pi f \tau)$$

$$T_s = 1/f_s = 50 \text{ ns}, \tau = 25 \text{ ns}$$

$$H(f) = \text{sinc}(\pi f T_s)$$

$$H(f) = \text{sinc}(\pi f T_s) = \sin(\pi f T_s) / (\pi f T_s) = \sin(\pi f / f_s) / (\pi f / f_s)$$

$$|H(f)| = |\sin(\pi f / f_s) / (\pi f / f_s)|$$

$$\text{In dB, } |H(f)| = 20 \log |\sin(\pi f / f_s) / (\pi f / f_s)|$$

$$|H(7 \text{ MHz})| = -1.83 \text{ dB}$$

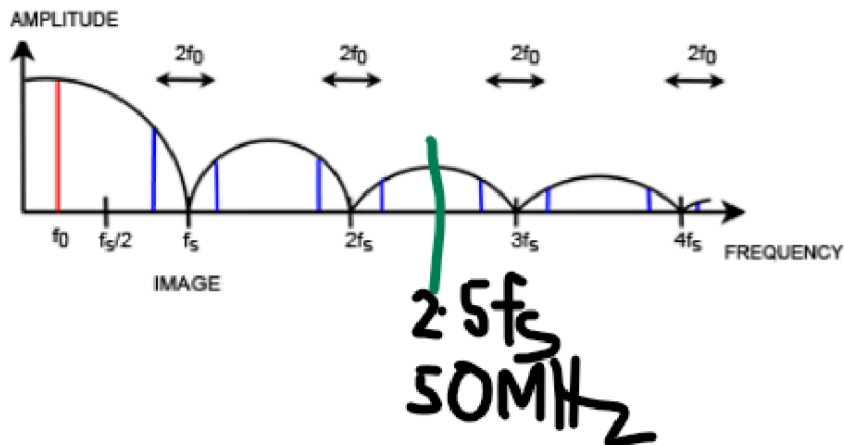
$$|H(13 \text{ MHz})| = -7.2 \text{ dB}$$

$$|H(27 \text{ MHz})| = -13.6 \text{ dB}$$

$$|H(33 \text{ MHz})| = -15.3 \text{ dB}$$

$$|H(47 \text{ MHz})| = -18.4 \text{ dB}$$

(b) Sketch the DAC output frequency spectrum up to $f = 50 \text{ MHz}$

(4 marks)

$$2.5f_s = 50 \text{ MHz}$$

- a) A circuit, shown in Fig. 2, is proposed to build an analog filter.
 (i) Derive the transfer-function $V_o(s)/V_i(s)$. What type of filter (bilinear, biquad or butterworth) is it?

KCL equations

- (1) $V_i/R_1 + V_2 sC - V_o (1/R + sC_1) = 0 \Rightarrow V_2 sC - V_o (1/R + sC_1) = -V_i/R_1$
 (2) $V_2/R_1 + V_o sC = 0 \Rightarrow V_2 = -V_o sC R_1$

Solve these 2 equations

- (1) $-V_o sC R_1 sC - V_o (1/R + sC_1) = -V_i/R_1 \Rightarrow V_o (1/R + sC_1 + (sC)^2 R_1) = V_i/R_1$
 $V_o/V_i = 1/[s^2(CR_1)^2 + sC_1 R_1 + R_1/R] = \{1/(CR_1)^2\}/[s^2 + sC_1/(C^2 R_1) + 1/(CR_1 CR)] \Rightarrow$
Biquad filter

(4 marks)

- (ii) For positive circuit elements, is the circuit *always* stable, unstable or depends on circuit values? Give your reasons.

$$\text{Filter poles } S_p = [-C_1/(C^2 R_1) \pm \sqrt{\{C_1/(C^2 R_1)\}^2 - 4/(CR_1 CR)}]/2$$

$$= [C_1/(2C^2 R_1)] [-1 \pm \sqrt{\{1 - 4/(C_1 R)\}}]$$

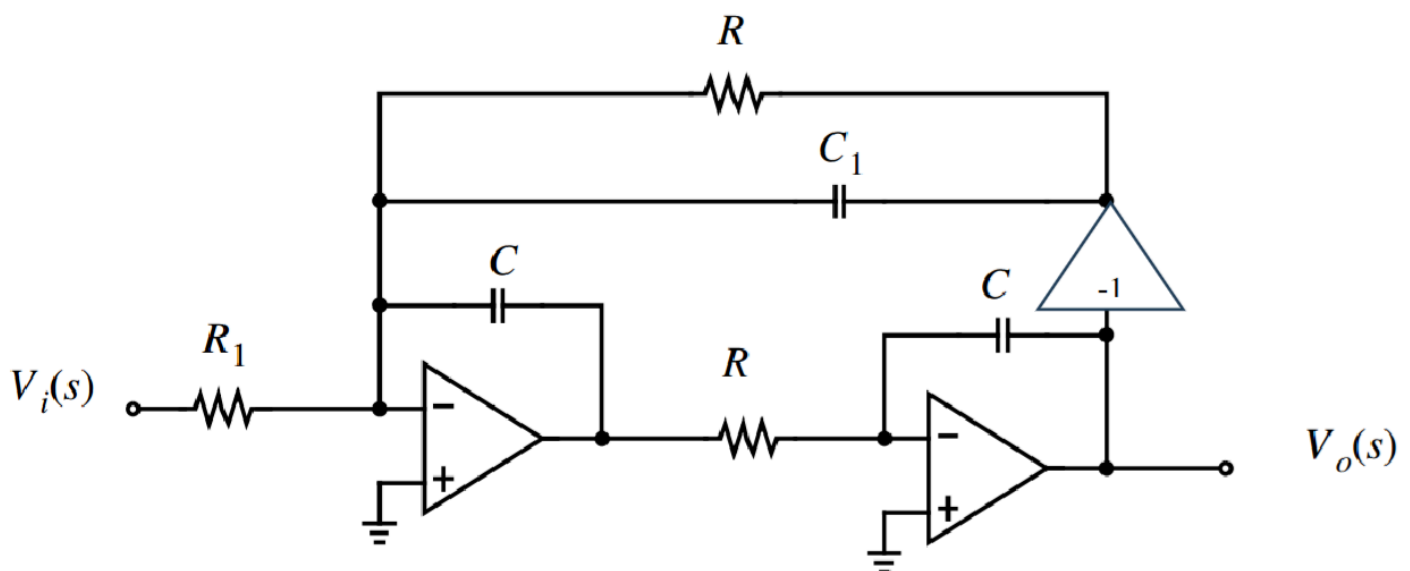
$$S_{p1} = [C_1/(2C^2 R_1)] [-1 + \sqrt{\{1 - 4/(C_1 R)\}}] \Rightarrow \text{LHP means stable}$$

$$S_{p2} = [C_1/(2C^2 R_1)] [-1 - \sqrt{\{1 - 4/(C_1 R)\}}] \Rightarrow \text{LHP means stable}$$

(2 marks)

- (iii) If unstable, how can the circuit be modified to make it stable. (4 marks)

Circuit is stable, so no fix needed.



- the transfer function a bit diff with solution

$$\begin{aligned}
 H_e(z) &= (1 - z^{-1})^2 \\
 H_e(j\omega) &= (1 - e^{-j\omega T})^2 = \left(2e^{-j\omega T/2} \left(\frac{e^{j\omega T/2} - e^{-j\omega T/2}}{2} \right) \right)^2 \\
 &= \left(2e^{-j\frac{\omega T}{2}} \left(j \sin\left(\frac{\omega T}{2}\right) \right) \right)^2 = \left(2 \sin\left(\frac{\omega T}{2}\right) e^{-j\frac{\omega T - \pi}{2}} \right)^2
 \end{aligned}$$

$$|H_e(f)| = \left(2 \sin(\pi f T) \right)^2 = \left(2 \sin\left(\pi \frac{f}{f_s}\right) \right)^2$$

(5 marks)

$$P_{\text{noise}} = \text{Integral}[\text{psd}(e) * |H_e(f)|^2]$$

Assuming $\omega T \ll 1$, so $\sin x \cong x$ for $x \ll 1$

$$\begin{aligned}
 P_{\text{noise}} &= \int_0^{f_B} \frac{\Delta^2}{12} \cdot \frac{2}{f_s} \cdot \left[2 \sin\left(\pi \frac{f}{f_s}\right) \right]^4 df \\
 &\cong \int_0^{f_B} \frac{\Delta^2}{12} \cdot \frac{2}{f_s} \cdot \left[2\pi \frac{f}{f_s} \right]^4 df \\
 &\cong \frac{\Delta^2}{12} \cdot \frac{\pi^4}{5} \left[\frac{2f_B}{f_s} \right]^5 = \frac{\Delta^2}{12} \cdot \frac{\pi^4}{5} \frac{1}{M^5}
 \end{aligned}$$

(5 marks)